

| No. | Project Name               | Project Short Description                                                                                                                                                                                                                                                                                              | Project Type                   | Target Department       |
|-----|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------|
| 1   | HVDC cable reverse voltage | The HVDC Cable Polarity Reversal Simulation is a highly coupled electro-thermal modeling initiative designed to predict dielectric breakdown inside polymer insulation during transient grid events. this project utilizes advanced FEA to digitally simulate the injection, migration, and trapping of space charges. | Hard<br><b>(Researchers)</b>   | Electrical Engineering  |
| 2   | Camera Defect Detection    | This industrial automation system identifies and classifies surface defects on continuously extruded power cables in real-time. Utilizing a 360-degree camera array and a custom-trained computer vision model, it categorizes defect type and severity by simple GUI.                                                 | Medium<br><b>(Researchers)</b> | Electronics Engineering |
| 3   | RFID Drum tracking         | Software System provides automated, real-time visibility of material movement and inventory state across the manufacturing floor. Utilizing ruggedized RFID tags and strategically placed reader portals. Crucially, the system connects with ORACLE.                                                                  | Simple<br><b>(Students)</b>    | Electronics Engineering |
| 4   | PPE Detection              | AI computer vision platform that simultaneously monitors live video streams for worker safety compliance (PPE) and facility-level threats (e.g.,                                                                                                                                                                       | Simple<br><b>(Students)</b>    | Electronics Engineering |

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|   |                         | smoke, fire, oil spills). By consolidating human and environmental monitoring into a single architectural pipeline. The structural of this deployment relies on deploying GPU compute to handle multi-model processing and tuning.                                                                                                                       |                              |                          |
| 5 | SOS Band                | Custom-manufactured safety tracking platform designed for high-precision worker localization. provide sub-meter tracking accuracy, instant manual SOS alerting, and automated fall detection. the system depends on maintaining dense, anchor coverage across the factory floor and ensuring the internally fabricated battery and enclosure assemblies. | Simple<br><b>(Students)</b>  | Electronics Engineering  |
| 6 | AI Agent                | The Enterprise AI Agent is a localized and designed to automate complex data querying and analysis across Quality Control, the Technical Office, or Finance. By securely indexing isolated enterprise data. the agent provides verifiable, context-aware answers to departmental queries.                                                                | Simple<br><b>(Students)</b>  | Computer Science         |
| 7 | Industrial Digital Twin | Dynamic, data-driven virtual replica of the physical manufacturing, powered by continuous IoT and PLC telemetry. it functions as an active simulation engine designed to predict equipment failures,                                                                                                                                                     | Hard<br><b>(Researchers)</b> | Mechatronics Engineering |

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|           |                                               | identify bottlenecks, and validate operational changes without risking physical production.                                                                                                                                                                                                                                            |                                |                                                  |
| <b>8</b>  | Autonomous Forklift                           | Manufacture local automated forklift design to carry heavy materials or drums                                                                                                                                                                                                                                                          | Medium<br><b>(Researchers)</b> | Mechatronics Engineering                         |
| <b>9</b>  | AI predictive maintenance                     | Condition-monitoring architecture that analyzes high-frequency, time-series telemetry—primarily vibration, acoustics, and temperature—to forecast mechanical degradation before catastrophic failure. the system utilizes machine learning anomaly detection to dynamically optimize asset uptime and spare parts inventory.           | Hard<br><b>(Researchers)</b>   | Mechanical Engineering                           |
| <b>10</b> | General Simulation (Drum, Extruders, ...etc.) | Multiphysics Simulation initiative deploys advanced Computer-Aided Engineering (CAE) to mathematically model two critical manufacturing realities: the non-Newtonian fluid dynamics of polymer melts within the extruder, and the structural integrity of high-mass cable drums under dynamic winding loads, other simulation projects | Simple<br><b>(Students)</b>    | Mechanical Engineering                           |
| <b>11</b> | XLPE Recycle                                  | The XLPE Recovery and Recycling initiative is a specialized material processing system designed to reclaim thermoset polymer waste generated during                                                                                                                                                                                    | Hard<br><b>(Researchers)</b>   | Metallurgical Engineering + Chemistry Department |

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|    |                             | power cable extrusion, this process utilizes advanced treatment to convert scrap into viable filler material or lower-grade industrial compounds.                                                                                                                                                                                  |                              |                                                  |
| 12 | New Material Simulation     | The Materials Formulation and Recipe Simulation platform utilizes materials informatics and computational modeling to accelerate the R&D of new polymer blends, composites, and extrusion compounds. the system optimizes additive ratios for target variables such as thermal stability, dielectric strength, and cost-reduction. | Hard<br><b>(Researchers)</b> | Metallurgical Engineering + Chemistry Department |
| 13 | Spray nano coating          | To make good heat dissipation of conductor surface can carry more current for the same cross section area                                                                                                                                                                                                                          | Hard<br><b>(Researchers)</b> | Metallurgical Engineering + Chemistry Department |
| 14 | Increase Metal Conductivity | The project aims to push conductivity limits beyond standard industry baselines.                                                                                                                                                                                                                                                   | Hard<br><b>(Researchers)</b> | Metallurgical Engineering + Faculty of Science   |
| 15 | Heat dissipation for OHTL   | maximizing thermal radiation away from the cable and reflecting incident solar energy, the coating lowers the conductor's operating temperature, thereby reducing thermal sag and unlocking additional ampacity on existing infrastructure                                                                                         | Hard<br><b>(Researchers)</b> | Metallurgical Engineering + Faculty of Science   |

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| <b>16</b> | Polymers for Cable Manufacturing | to know more about XLPE, PVC, LSHF, PE, Rubber polymers behavior | Simple<br><b>(Students)</b> | Metallurgical Engineering + Chemistry Department |
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